PhD Topology Exam
January 2026

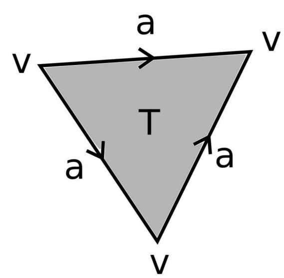
Be neat, do not write too small, and leave margins.

For the first five problems, show all of your work and support all statements. Use a separate sheet of paper for each problem. (Each problem 10 points.)

1. Prove that every metrizable space (X, d) is normal.

2. Let (X, d) be a compact metric space. Let $f\colon X \to X$ and suppose there is some $c < 1$ with $d(f(x), f(y)) \leq c \cdot d(x, y)$ for all $x, y \in X$. Prove there exists a unique $x \in X$ with $f(x) = x$.

3. Consider the Δ-complex X drawn below. (Note the orientation of the three edges carefully.) Compute the simplicial homology $H_i(X)$ for $i = 0, 1, 2$.



4. For M a compact orientable odd-dimensional manifold without boundary, show that the Euler characteristic $\chi(M)$ is zero.

5. (a) Let $n > m$. Show there are no maps from $\mathbb{RP}^n$ to $\mathbb{RP}^m$ inducing a nontrivial map $H^1(\mathbb{RP}^m; \mathbb{Z}_2) \to H^1(\mathbb{RP}^n; \mathbb{Z}_2)$.
   (b) Prove the Borsuk–Ulam theorem as follows. Suppose on the contrary that $f\colon S^n \to \mathbb{R}^n$ satisfies $f(x) \neq f(-x)$ for all x. Define $g\colon S^n \to S^{n-1}$ by $g(x) = \frac{f(x)-f(-x)}{\|f(x)-f(-x)\|}$, so $g(-x) = -g(x)$ and g induces a map $\mathbb{RP}^n \to \mathbb{RP}^{n-1}$. Show that (a) applies to this map.

The remaining problems are on the following page.

Answer the following ten problems with complete definitions, complete statements, an example, or a short proof. (Each problem 5 points.)

6. State the Tietze Extension Theorem.

7. Let M_g be the orientable surface of genus g, i.e., the g-fold connected sum of tori. Use the Seifert van-Kampen theorem to derive the fundamental group of M_g for $g \geq 1$.

8. Show that if $g\colon S^n \to S^n$ is continuous and $g(x) \neq g(-x)$ for all $x \in S^n$, then g is surjective.

9. Prove that if $f\colon S^n \to S^n$ has no fixed points, then $\deg(f) = (-1)^{n+1}$.

10. Show that $\mathbb{R}^\omega$ with the box topology is not connected.

11. State the Five-Lemma.

12. Show that every map $f\colon \mathbb{CP}^n \to \mathbb{CP}^n$ has a fixed point if n is even.

13. Let $p\colon Y \to X$ and $q\colon Z \to X$ be basepoint-preserving covering space maps, where X, Y, Z are path-connected and locally path-connected. Prove that if $p_*(\pi_1(Y)) = q_*(\pi_1(Z))$, then the covering spaces p and q are isomorphic.

14. Does there exist a closed orientable 18-dimensional manifold M with $H^9(M;\mathbb{Z}) \cong \mathbb{Z}$?

15. Prove that if X is the Möbius band and A is its boundary circle, then there is no retraction $r\colon X \to A$.